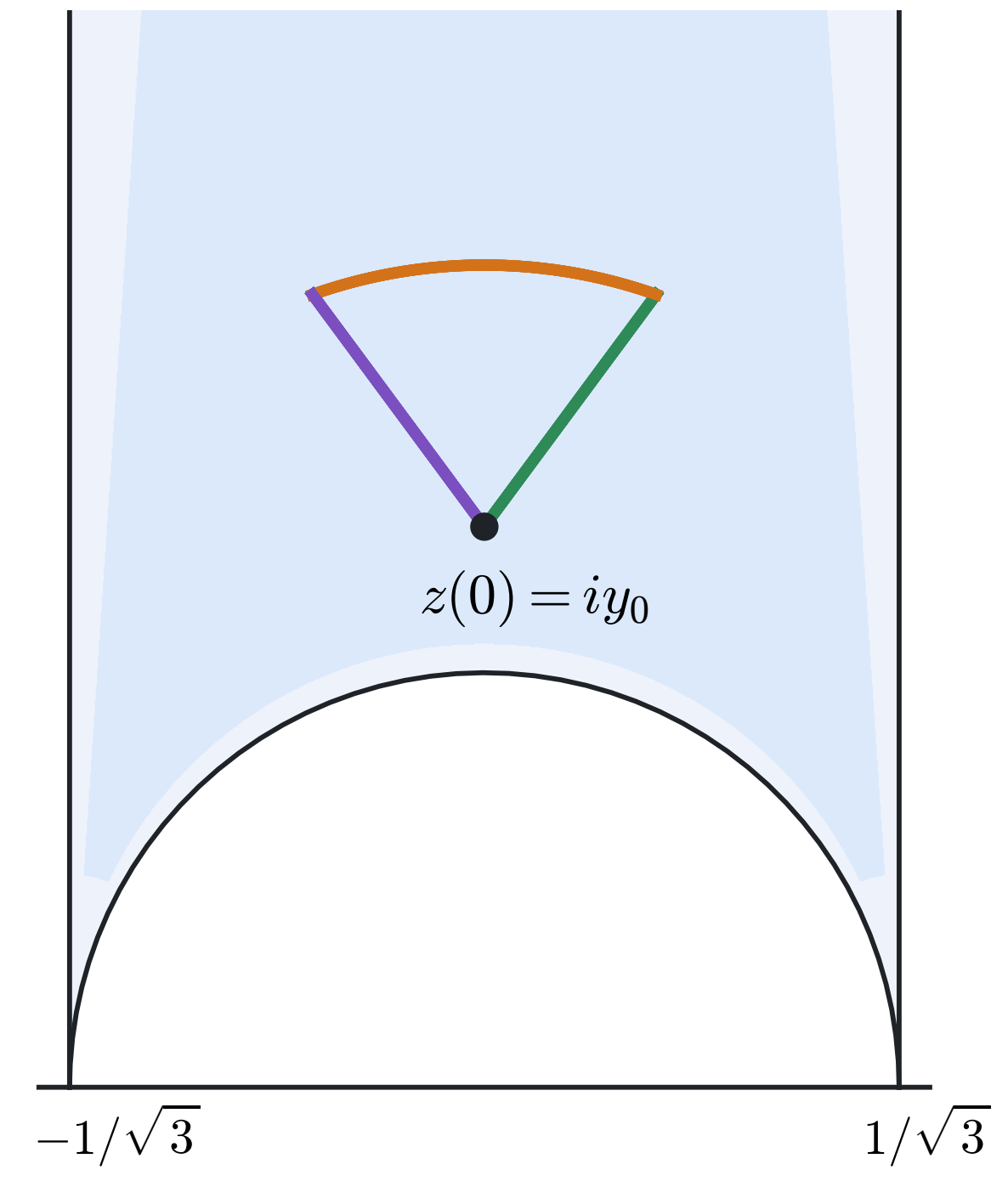

$z(0) = iy_0$
$-1/\sqrt{3}$
$1/\sqrt{3}$